

ASSIGNMENT NO. 2: Basic Data Retrieval using SQL

Questions:

Q1) Create a database named CollegeDB and a table named Student.

The table should contain the following columns:

Column Name	Data Type
StudentID	INT
StudentName	VARCHAR(50)
Age	INT
Course	VARCHAR(30)
City	VARCHAR(30)
Marks	INT

Add a primary key on StudentID and insert at least 10 records.

Query-

```
CREATE DATABASE CollegeDB;
```

```
USE CollegeDB;
```

```
CREATE TABLE Student (  
StudentID INT PRIMARY KEY,  
StudentName VARCHAR(50),  
Age INT,  
Course VARCHAR(30),  
City VARCHAR(30),  
Marks INT  
);
```

```
INSERT INTO Student (StudentID, StudentName, Age, Course, City, Marks) VALUES  
(1, 'Alice', 20, 'BSc', 'Mumbai', 85),  
(2, 'Bob', 21, 'BCA', 'Pune', 78),  
(3, 'Charlie', 19, 'BSc', 'Delhi', 92),  
(4, 'Diana', 22, 'MCA', 'Mumbai', 88),  
(5, 'Eve', 20, 'BCA', 'Nashik', 74),  
(6, 'Frank', 21, 'BSc', 'Mumbai', 65),  
(7, 'Grace', 19, 'MCA', 'Pune', 95),  
(8, 'Heidi', 22, 'BCA', 'Delhi', 81),  
(9, 'Ivan', 20, 'BSc', 'Mumbai', 89),  
(10, 'Judy', 21, 'MCA', 'Nashik', 76);
```

Output-

Showing rows 0 - 9 (10 total, Query took 0.0007 seconds.)

```
SELECT * FROM `student`
```

Profiling [Edit Inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	StudentID	StudentName	Age	Course	City	Marks
☐ Edit ☐ Copy ☐ Delete	1	Alice	20	BSc	Mumbai	85
☐ Edit ☐ Copy ☐ Delete	2	Bob	21	BCA	Pune	78
☐ Edit ☐ Copy ☐ Delete	3	Charlie	19	BSc	Delhi	92
☐ Edit ☐ Copy ☐ Delete	4	Diana	22	MCA	Mumbai	88
☐ Edit ☐ Copy ☐ Delete	5	Eve	20	BCA	Nashik	74
☐ Edit ☐ Copy ☐ Delete	6	Frank	21	BSc	Mumbai	65
☐ Edit ☐ Copy ☐ Delete	7	Grace	19	MCA	Pune	95
☐ Edit ☐ Copy ☐ Delete	8	Heidi	22	BCA	Delhi	81
☐ Edit ☐ Copy ☐ Delete	9	Ivan	20	BSc	Mumbai	89
☐ Edit ☐ Copy ☐ Delete	10	Judy	21	MCA	Nashik	76

Q2) Write a MySQL query to display all records from the Student table.
The output should display all columns and all rows.

Query-

SELECT * FROM Student;

Output-

Showing rows 0 - 9 (10 total, Query took 0.0015 seconds.)

```
SELECT * FROM Student;
```

Profiling [Edit Inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	StudentID	StudentName	Age	Course	City	Marks
☐ Edit ☐ Copy ☐ Delete	1	Alice	20	BSc	Mumbai	85
☐ Edit ☐ Copy ☐ Delete	2	Bob	21	BCA	Pune	78
☐ Edit ☐ Copy ☐ Delete	3	Charlie	19	BSc	Delhi	92
☐ Edit ☐ Copy ☐ Delete	4	Diana	22	MCA	Mumbai	88
☐ Edit ☐ Copy ☐ Delete	5	Eve	20	BCA	Nashik	74
☐ Edit ☐ Copy ☐ Delete	6	Frank	21	BSc	Mumbai	65
☐ Edit ☐ Copy ☐ Delete	7	Grace	19	MCA	Pune	95
☐ Edit ☐ Copy ☐ Delete	8	Heidi	22	BCA	Delhi	81
☐ Edit ☐ Copy ☐ Delete	9	Ivan	20	BSc	Mumbai	89
☐ Edit ☐ Copy ☐ Delete	10	Judy	21	MCA	Nashik	76

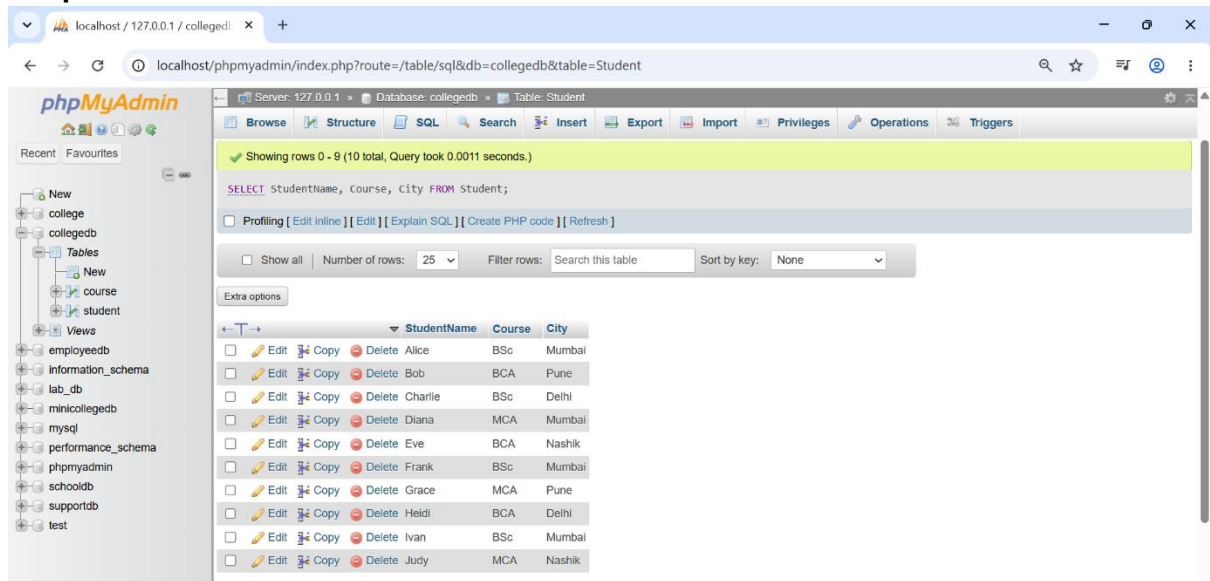
Q3) Write a MySQL query to display only the following columns:

1. StudentName
2. Course
3. City

Query-

SELECT StudentName, Course, City
FROM Student;

Output-



The screenshot shows the phpMyAdmin interface. The left sidebar displays the database structure, including 'collegedb' and its tables 'course' and 'student'. The main panel shows the 'Student' table with 10 rows. The SQL query executed is `SELECT StudentName, Course, City FROM Student;`. The table data is as follows:

StudentName	Course	City
Alice	BSc	Mumbai
Bob	BCA	Pune
Charlie	BSc	Delhi
Diana	MCA	Mumbai
Eve	BCA	Nashik
Frank	BSc	Mumbai
Grace	MCA	Pune
Heidi	BCA	Delhi
Ivan	BSc	Mumbai
Judy	MCA	Nashik

Q4) Write a MySQL query to display students who belong to the city Mumbai.

The output should include:

1. StudentID
2. StudentName
3. City
4. Marks

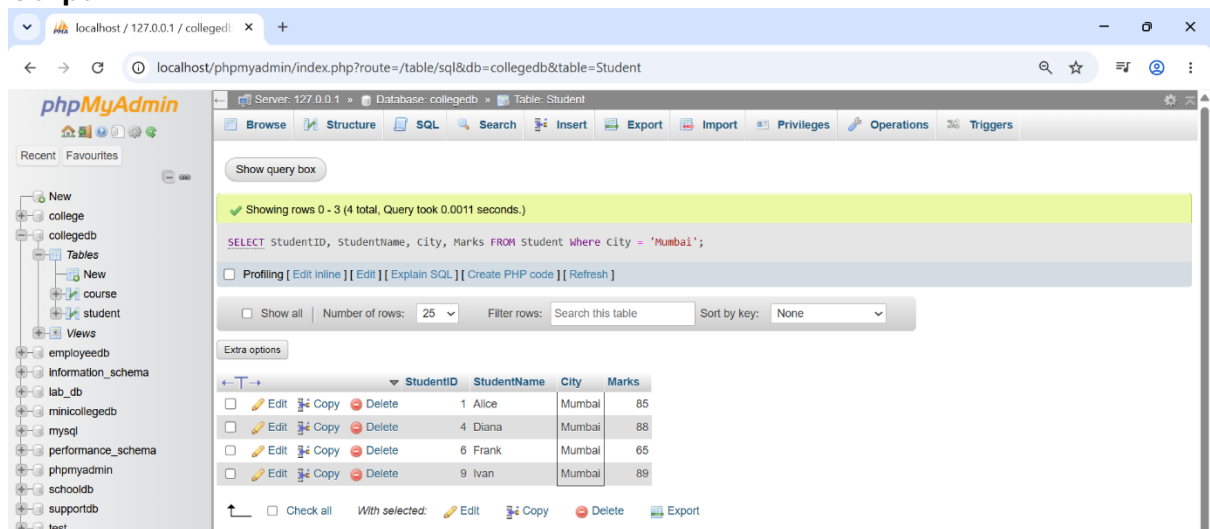
Query-

SELECT StudentID, StudentName, City, Marks

FROM Student

Where City = 'Mumbai';

Output-



The screenshot shows the phpMyAdmin interface. The left sidebar displays the database structure, including 'collegedb' and its tables 'course' and 'student'. The main panel shows the 'Student' table with 4 rows. The SQL query executed is `SELECT StudentID, StudentName, City, Marks FROM Student Where City = 'Mumbai';`. The table data is as follows:

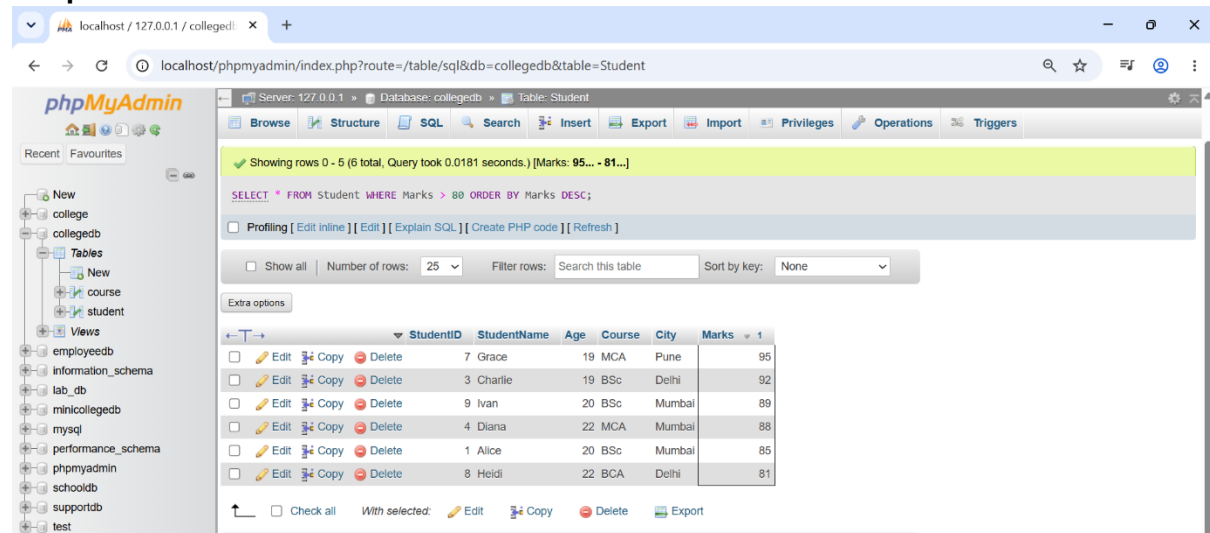
StudentID	StudentName	City	Marks
1	Alice	Mumbai	85
4	Diana	Mumbai	88
6	Frank	Mumbai	65
9	Ivan	Mumbai	89

Q5) Write a MySQL query to display students whose marks are greater than 80. The output should be sorted by marks in descending order.

Query-

```
SELECT *  
FROM Student  
WHERE Marks > 80  
ORDER BY Marks DESC;
```

Output-



Showing rows 0 - 5 (6 total, Query took 0.0181 seconds.) [Marks: 95... - 81...]

```
SELECT * FROM Student WHERE Marks > 80 ORDER BY Marks DESC;
```

Number of rows: 25 Filter rows: Search this table Sort by key: None

	StudentID	StudentName	Age	Course	City	Marks
☐	7	Grace	19	MCA	Pune	95
☐	3	Charlie	19	BSc	Delhi	92
☐	9	Ivan	20	BSc	Mumbai	89
☐	4	Diana	22	MCA	Mumbai	88
☐	1	Alice	20	BSc	Mumbai	85
☐	8	Heidi	22	BCA	Delhi	81

Q6) Write a MySQL query to display the top 3 students based on marks. The query should:

1. Display StudentName and Marks.
2. Sort marks from highest to lowest.
3. Display only 3 records using LIMIT.

Query-

```
SELECT StudentName, Marks  
FROM Student  
ORDER BY Marks DESC  
LIMIT 3;
```

Output-

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/table/sql&db=collegedb&table=Student

Server: 127.0.0.1 > Database: collegedb > Table: Student

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show query box

Showing rows 0 - 2 (3 total, Query took 0.0014 seconds.) [Marks: 95... - 89...]

```
SELECT StudentName, Marks FROM Student ORDER BY Marks DESC LIMIT 3;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

Extra options

	StudentName	Marks
☐ Edit Copy Delete	Grace	95
☐ Edit Copy Delete	Charlie	92
☐ Edit Copy Delete	Ivan	89

☐ Check all With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

Query results operations